Case Study: Application to AES MixColumns

Supplementary note to "Optimizing Polynomial Multiplication and Fixed-Weight Sampling for HQC on ARM Cortex-M4", IACR TCHES, Volume 2027, Issue 1. Jihoon Jang, Hanbeom Shin, Suhri Kim, Seokhie Hong, and Donggeun Kwon. This note reproduces the case study that was omitted from the published version for reasons of space.

The framework of Section 3 of the paper is not specific to the HQC butterfly but applies to any $\mathbb{F}_2$-linear layer given as an s-XOR sequence. To confirm this, we applied the same optimization and the same cost model ($K = 14$) to a 92-XOR s-XOR sequence of Xiang et al.[1] implementing the 32×32 binary matrix of AES MixColumns. The results are shown in Table 1: the input order exhibits the same pattern as Observation 1 of Section 3.2.1 of the paper, with VMOV instructions outnumbering EOR instructions, and after XOR reordering the VMOV count drops from 128 to 68 (-46.9%) and the total instruction count from 220 to 160 (-27.3%).

The full instruction listing of the resulting implementation is given in Table 2, and a baseline implementation that emits the input sequence in its given order (220 instructions) is given in Table 3. In both listings, `ld` x_i is a load VMOV that brings x_i from its VFP register into a GP register, and `sp` x_i is a spill VMOV that writes the latest value of a dirty x_i back to a VFP register; together they make up the VMOV counts of Table 1. The remaining entries are the in-place XORs of the sequence, one EOR instruction each, and $[y_j]$ marks the operation that produces output y_j. Evicting a clean variable requires no instruction and therefore appears in neither listing.

Table 1: Result of applying XOR reordering to a 92-XOR sequence for AES MixColumns.

	#EOR	#VMOV	Total
Input sequence	92	128	220
XOR reordering	92	68	160
Δ	—	-46.9%	-27.3%

[1] Zejun Xiang, Xiangyoung Zeng, Da Lin, Zhenzhen Bao, and Shasha Zhang. Optimizing implementations of linear layers. *IACR Transactions on Symmetric Cryptology*, 2020(2):120–145, 2020.

Table 2: Implementation of the AES MixColumns 92-XOR sequence after XOR reordering (160 instructions). `load`/`spill` denote load/spill VMOV instructions between GP and VFP registers.

No.	Operation	No.	Operation	No.	Operation	No.	Operation	No.	Operation
0	load x21	32	load x20	64	x11=x11⊕x10	96	spill x10	128	x7=x7⊕x23[y15]
1	load x13	33	x12=x12⊕x20	65	x11=x11⊕x15	97	load x16	129	spill x25
2	x13=x13⊕x21	34	spill x5	66	x19=x19⊕x27[y11]	98	x24=x24⊕x16	130	load x30
3	load x27	35	load x15	67	spill x19	99	x16=x16⊕x31	131	x14=x14⊕x30
4	load x11	36	x31=x31⊕x15	68	load x2	100	x17=x17⊕x16	132	spill x7
5	x11=x11⊕x27	37	spill x14	69	x11=x11⊕x2[y3]	101	x12=x12⊕x15[y28]	133	load x21
6	load x6	38	load x28	70	x10=x10⊕x18	102	spill x2	134	x14=x14⊕x21[y6]
7	load x14	39	x4=x4⊕x28	71	x27=x27⊕x11[y27]	103	load x8	135	x6=x6⊕x14
8	x14=x14⊕x6	40	x29=x29⊕x4	72	spill x27	104	x24=x24⊕x8	136	x22=x22⊕x21
9	load x31	41	x4=x4⊕x11	73	load x9	105	x16=x16⊕x0	137	spill x14
10	load x23	42	spill x29	74	spill x11	106	x25=x25⊕x24	138	load x28
11	x23=x23⊕x31	43	load x7	75	load x17	107	x8=x8⊕x23	139	x21=x21⊕x28
12	load x22	44	x15=x15⊕x7	76	x17=x17⊕x9	108	x9=x9⊕x8	140	load x5
13	x6=x6⊕x22	45	load x19	77	x18=x18⊕x17	109	x4=x4⊕x31	141	x6=x6⊕x5[y22]
14	load x4	46	x11=x11⊕x19	78	spill x3	110	x8=x8⊕x0[y16]	142	x23=x23⊕x31[y31]
15	load x12	47	x30=x30⊕x6[y14]	79	load x25	111	x0=x0⊕x31	143	spill x31
16	x12=x12⊕x4	48	x19=x19⊕x23	80	x17=x17⊕x25	112	x24=x24⊕x15[y0]	144	load x29
17	load x30	49	x28=x28⊕x12	81	spill x30	113	x15=x15⊕x23	145	x5=x5⊕x29[y29]
18	x22=x22⊕x30	50	x20=x20⊕x27	82	load x1	114	x16=x16⊕x23[y8]	146	x9=x9⊕x17[y9]
19	load x5	51	spill x28	83	x9=x9⊕x1	115	x1=x1⊕x25	147	x29=x29⊕x21[y13]
20	x21=x21⊕x5	52	load x3	84	x10=x10⊕x9[y2]	116	spill x16	148	x0=x0⊕x24[y24]
21	load x26	53	x27=x27⊕x3	85	x1=x1⊕x17	117	load x7	149	spill x23
22	spill x21	54	x20=x20⊕x19	86	x18=x18⊕x2[y10]	118	x31=x31⊕x7	150	load x20
23	load x18	55	x27=x27⊕x31	87	x26=x26⊕x1	119	spill x8	151	x28=x28⊕x20[y20]
24	x18=x18⊕x26	56	x12=x12⊕x27	88	x2=x2⊕x26	120	load x14	152	x20=x20⊕x4[y12]
25	x30=x30⊕x13	57	x19=x19⊕x18	89	x26=x26⊕x18[y26]	121	x7=x7⊕x14	153	x4=x4⊕x12[y4]
26	spill x22	58	spill x20	90	x2=x2⊕x9[y18]	122	x25=x25⊕x9[y17]	154	x17=x17⊕x0[y1]
27	load x29	59	load x10	91	spill x26	123	x15=x15⊕x7[y7]	155	x1=x1⊕x0[y25]
28	x13=x13⊕x29	60	x26=x26⊕x10	92	load x24	124	x23=x23⊕x6	156	spill x4
29	x5=x5⊕x13	61	x3=x3⊕x19	93	spill x18	125	spill x15	157	load x13
30	x13=x13⊕x12[y5]	62	x27=x27⊕x26	94	load x0	126	load x22	158	x21=x21⊕x13[y21]
31	spill x13	63	x3=x3⊕x11[y19]	95	x0=x0⊕x24	127	x31=x31⊕x22[y23]	159	x22=x22⊕x6[y30]

Table 3: Baseline implementation of the AES MixColumns 92-XOR sequence, emitted in the given order (220 instructions). `load`/`spill` denote load/spill VMOV instructions between GP and VFP registers.

No.	Operation	No.	Operation	No.	Operation	No.	Operation	No.	Operation
0	load x31	44	x24=x24⊕x8	88	load x18	132	spill x17	176	spill x30
1	load x23	45	spill x16	89	spill x12	133	load x7	177	load x19
2	x23=x23⊕x31	46	load x26	90	load x19	134	x31=x31⊕x7	178	spill x12
3	load x15	47	spill x24	91	x19=x19⊕x18	135	spill x31	179	load x3
4	x31=x31⊕x15	48	load x18	92	x11=x11⊕x10	136	load x22	180	x3=x3⊕x19[y19]
5	load x4	49	x18=x18⊕x26	93	x10=x10⊕x18	137	spill x23	181	x19=x19⊕x27[y11]
6	load x12	50	load x30	94	x18=x18⊕x2	138	load x31	182	x1=x1⊕x25
7	x12=x12⊕x4	51	spill x18	95	spill x10	139	x31=x31⊕x22[y23]	183	x0=x0⊕x24[y24]
8	load x21	52	load x22	96	load x9	140	x30=x30⊕x6[y14]	184	x1=x1⊕x0[y25]
9	load x13	53	x22=x22⊕x30	97	spill x19	141	spill x31	185	spill x24
10	x13=x13⊕x21	54	load x10	98	load x10	142	load x14	186	load x2
11	load x9	55	spill x22	99	x10=x10⊕x9[y2]	143	x7=x7⊕x14	187	x2=x2⊕x26[y18]
12	load x17	56	load x26	100	x2=x2⊕x9	144	spill x15	188	x25=x25⊕x9[y17]
13	x17=x17⊕x9	57	x26=x26⊕x10	101	spill x10	145	load x21	189	spill x25
14	load x27	58	x8=x8⊕x23	102	load x17	146	x14=x14⊕x21	190	load x15
15	load x11	59	x30=x30⊕x13	103	x18=x18⊕x17[y10]	147	x6=x6⊕x5	191	x15=x15⊕x7[y7]
16	x11=x11⊕x27	60	spill x26	104	spill x18	148	x22=x22⊕x21	192	spill x7
17	load x28	61	load x29	105	load x25	149	spill x5	193	load x23
18	x4=x4⊕x28	62	x13=x13⊕x29	106	x17=x17⊕x25	150	load x29	194	spill x28
19	load x5	63	x5=x5⊕x13	107	spill x2	151	spill x22	195	load x7
20	x21=x21⊕x5	64	x29=x29⊕x4	108	load x1	152	load x5	196	x7=x7⊕x23[y15]
21	load x24	65	x4=x4⊕x11	109	x1=x1⊕x17	153	x5=x5⊕x29[y29]	197	x6=x6⊕x14[y22]
22	spill x21	66	spill x29	110	load x24	154	x21=x21⊕x28	198	spill x9
23	load x0	67	load x19	111	spill x1	155	x29=x29⊕x21[y13]	199	load x17
24	x0=x0⊕x24	68	x11=x11⊕x19	112	load x25	156	x21=x21⊕x13[y21]	200	spill x14
25	spill x0	69	x13=x13⊕x12[y5]	113	x25=x25⊕x24	157	spill x13	201	load x9
26	load x7	70	x19=x19⊕x23	114	x9=x9⊕x8	158	load x12	202	x9=x9⊕x17[y9]
27	x15=x15⊕x7	71	x4=x4⊕x31	115	x24=x24⊕x15[y0]	159	x12=x12⊕x27[y28]	203	spill x4
28	load x1	72	spill x4	116	x11=x11⊕x15[y3]	160	spill x21	204	load x31
29	x9=x9⊕x1	73	load x20	117	spill x8	161	load x26	205	x23=x23⊕x31[y31]
30	load x6	74	x12=x12⊕x20	118	load x0	162	x27=x27⊕x26	206	spill x26
31	spill x17	75	x28=x28⊕x12	119	spill x11	163	spill x28	207	load x18
32	load x14	76	x20=x20⊕x27	120	load x8	164	load x20	208	spill x1
33	x14=x14⊕x6	77	x20=x20⊕x19	121	x8=x8⊕x0[y16]	165	spill x29	209	load x26
34	spill x14	78	x27=x27⊕x31	122	x15=x15⊕x23	166	load x28	210	x26=x26⊕x18[y26]
35	load x16	79	x12=x12⊕x15	123	spill x8	167	x28=x28⊕x20[y20]	211	spill x19
36	x24=x24⊕x16	80	spill x20	124	load x16	168	spill x5	212	load x22
37	load x22	81	load x3	125	x17=x17⊕x16	169	load x4	213	x22=x22⊕x6[y30]
38	spill x9	82	x27=x27⊕x3	126	x16=x16⊕x0	170	x20=x20⊕x4[y12]	214	x17=x17⊕x0[y1]
39	load x6	83	x3=x3⊕x11	127	x0=x0⊕x31	171	spill x20	215	spill x27
40	x6=x6⊕x22	84	spill x3	128	x16=x16⊕x23[y8]	172	load x1	216	load x11
41	x16=x16⊕x31	85	load x2	129	spill x16	173	x26=x26⊕x1	217	spill x0
42	spill x6	86	x11=x11⊕x2	130	load x6	174	x14=x14⊕x30[y6]	218	load x27
43	load x8	87	spill x19	131	x23=x23⊕x6	175	x4=x4⊕x12[y4]	219	x27=x27⊕x11[y27]